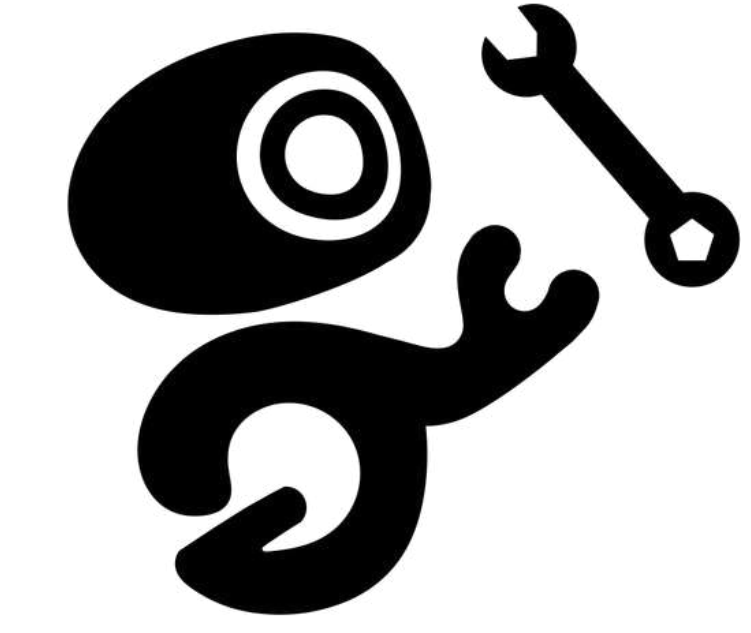


Dynamic Symmetry for Orientation-Independent Planetary Mobility

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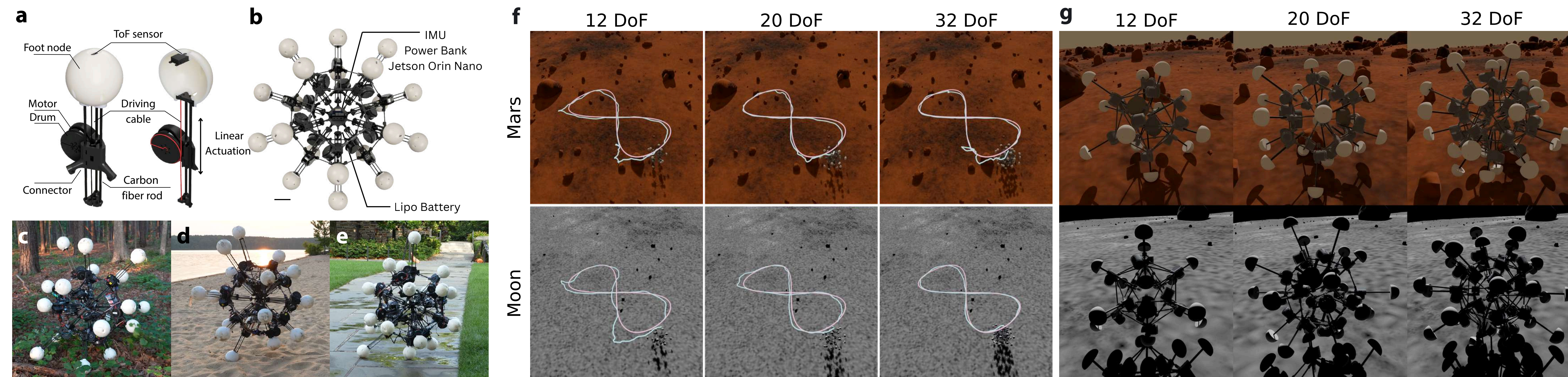


General Robotics Lab



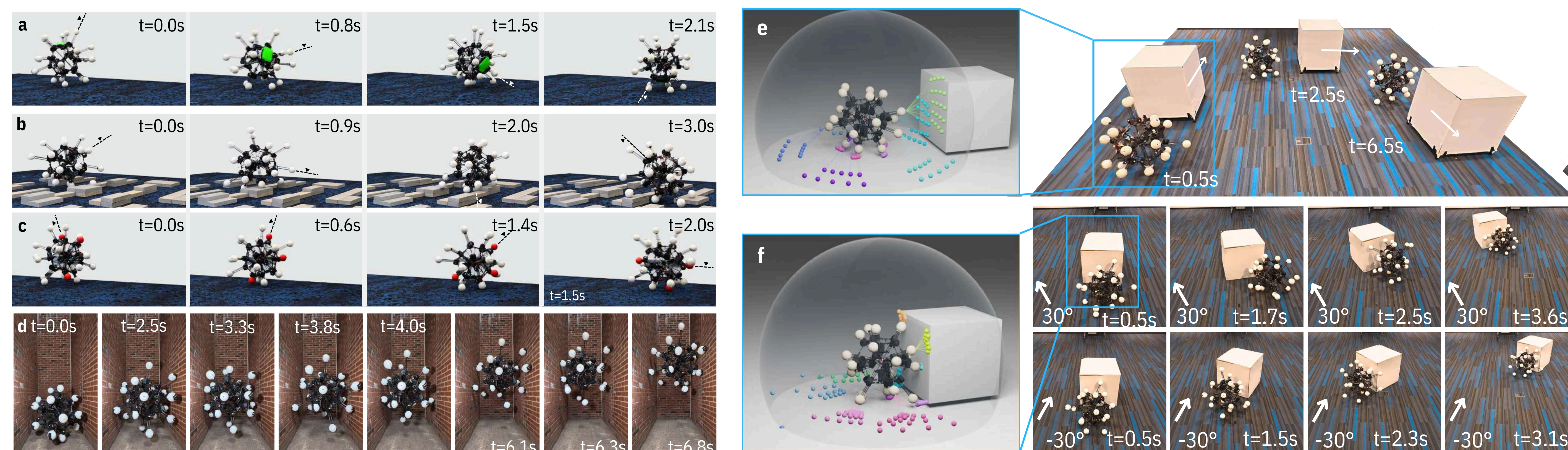
Website

Argus platform[1], real-world locomotion and simulated planetary exploration.



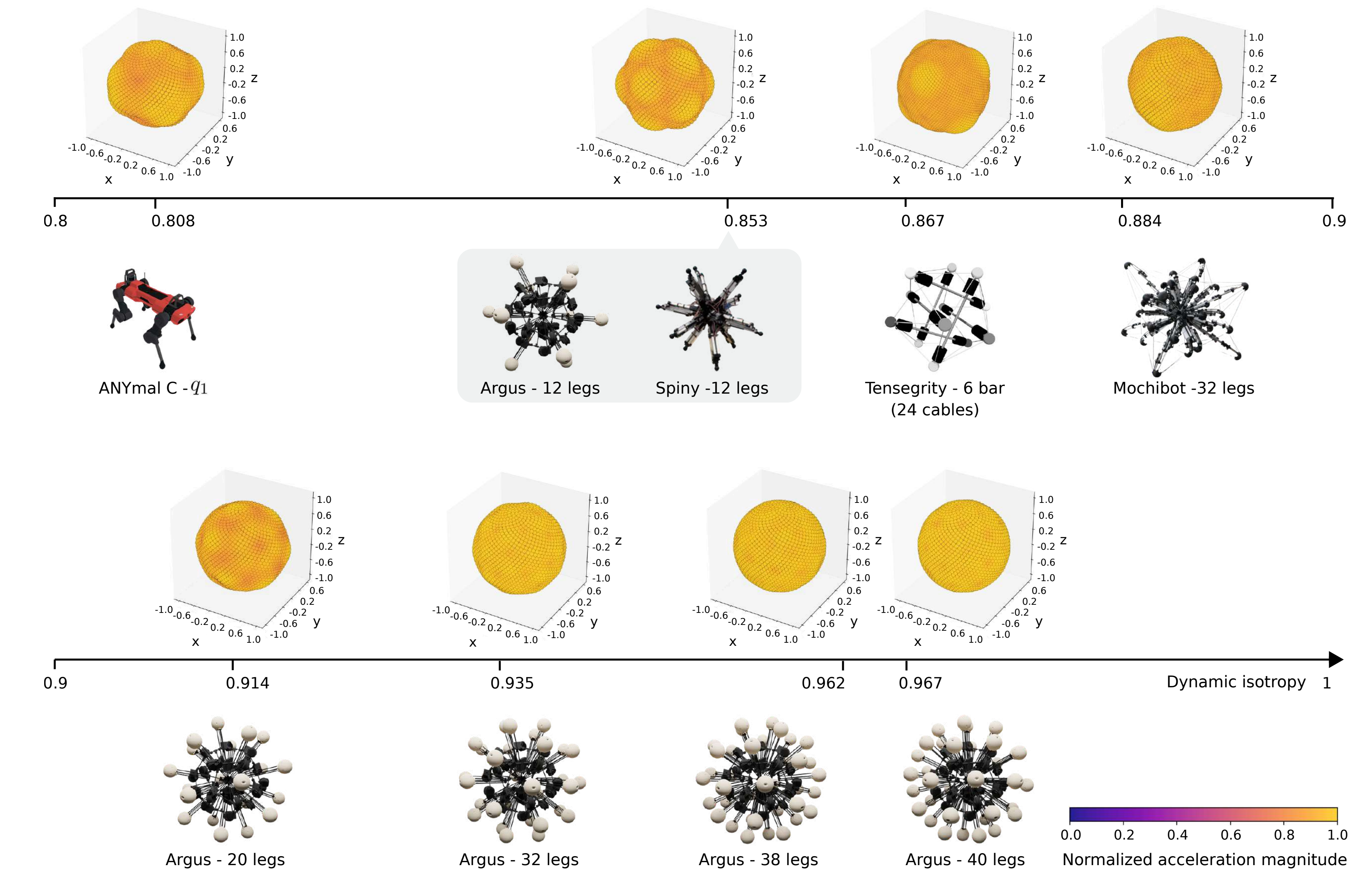
(a) Argus leg module with a radially oriented 1-degree-of-freedom (DoF) cable-driven linear actuator and an time-of-flight (ToF) depth sensor. (b) The 20-leg dodecahedral Argus robot with near-extreme dynamic isotropy ($\eta = 0.91$). Scale bar, 0.1 m. (c-e) The flat-ground policy transfers zero-shot to unstructured outdoor terrain: dense foliage with high vegetation (c), soft sand (d), and grass and pavement (e). (f-g) Argus with 12/20/32 DoF following a figure-eight trajectory on simulated Mars and Moon terrain [2] with respective gravity. (f) Fixed top-down camera view with overlaid commanded (pink) path and actual (cyan) trajectory. (g) Camera view following the robot.

Real world demonstration of the mobility, load-bearing, robustness, full coverage of perception and whole body manipulation with the 20-leg Argus.

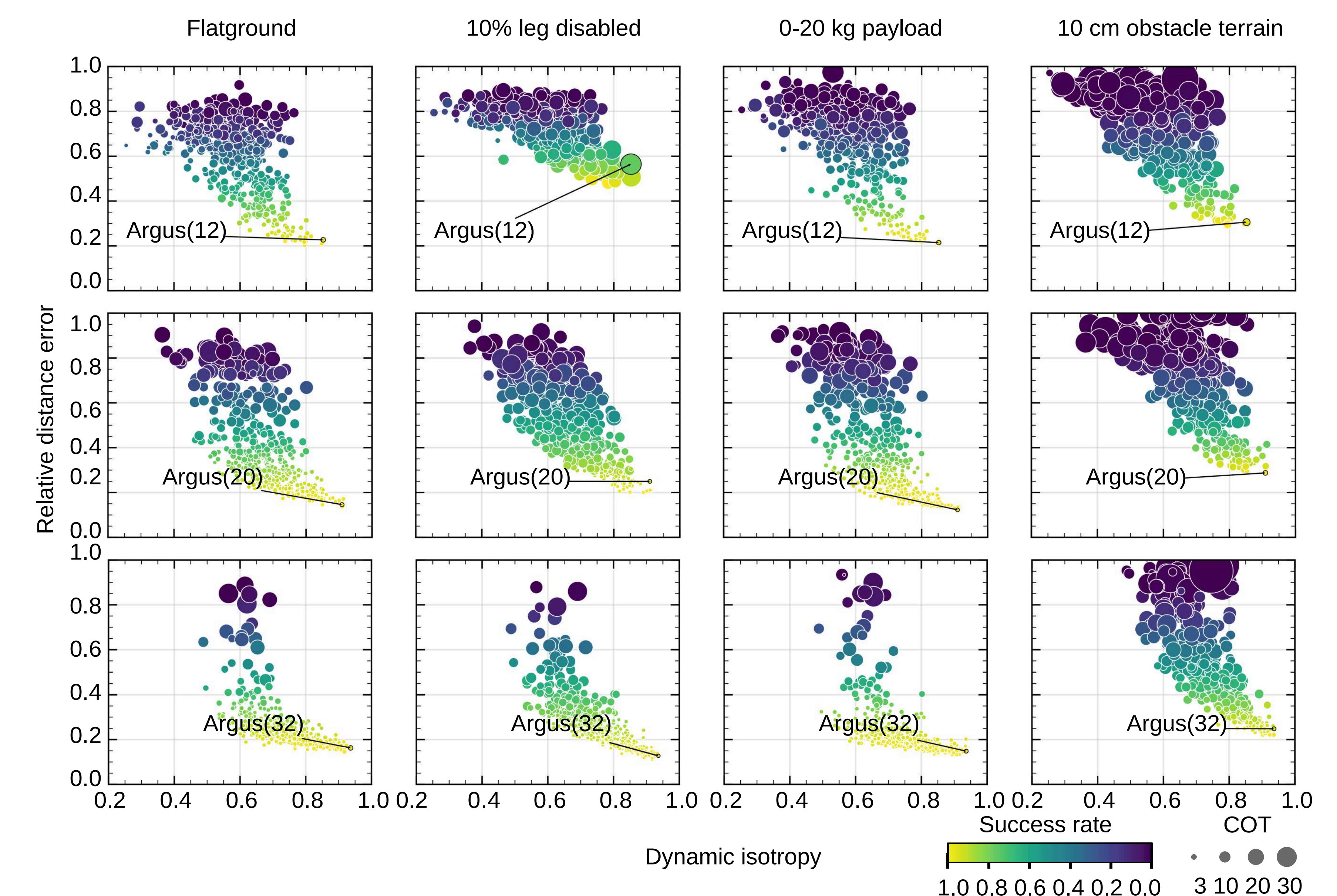


(a) Argus rolls with an asymmetrically mounted payload (green). (b) It reconfigures contact legs to traverse discrete terrain. (c) With one to three disabled legs (red), it reorients to bring healthy legs into ground contact. (d) Under counterweight-emulated lunar gravity, it climbs between parallel walls by alternating bracing and thrusting motions. (e) In simulation, fused depth measurements from 20 foot-mounted ToF cameras enable moving-target tracking. (f) In simulation, the estimated target state supports perception-guided whole-body pushing and following.

Dynamic isotropy measured by min/max CoM attainable acceleration



Dynamic isotropy vs Task performance



[1] J. Liu, B. Xia, and B. Chen, "Extreme dynamic symmetry enables omnidirectional and multifunctional robots," Science Robotics, 2026.

[2] A. Orsula, M. Geist, M. Olivares-Mendez, and C. Martinez, "Space robotics bench: robot learning beyond earth," arXiv preprint arXiv:2509.23328, 2025.